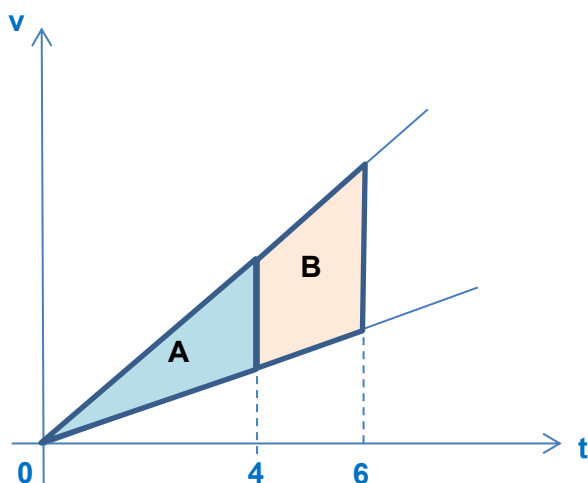


4	Two cars are initially side-by-side and at rest. They then accelerate in the same direction along the same straight line at different uniform rates. After 4.0 s, one of the cars is 12 m ahead of the other. If they continue to accelerate at the same rate, how far apart will they be 6.0 s after they started?							
	A	15 m	B	18 m	C	22 m	D	27 m

Answer: D

Method 1 (graphical method)



Distance apart at 4 s = Shaded area A = 12 m

Distance apart at 6 s = Shaded areas A and B

By similar triangles,

$$\frac{\text{Area A}}{\text{Area A} + \text{Area B}} = \left(\frac{4}{6}\right)^2$$

$$\frac{12}{\text{Area A} + \text{Area B}} = \left(\frac{4}{6}\right)^2$$

$$\text{Area A} + \text{Area B} = 12 \left(\frac{6}{4}\right)^2 = 27 \text{ m}$$

Method 2 (using kinematics equations)

Using $S = ut + \frac{1}{2}at^2$

Since they both start from rest, $u = 0$.

At $t = 4 \text{ s}$, $S_1 - S_2 = \frac{1}{2}a_1(4)^2 - \frac{1}{2}a_2(4)^2 = 12 \text{ m}$

Hence $a_1 - a_2 = 24/16$

At $t = 6 \text{ s}$, $S_1 - S_2 = \frac{1}{2}a_1(6)^2 - \frac{1}{2}a_2(6)^2 = \frac{1}{2}(6)^2(a_1 - a_2) = 27 \text{ m}$

5 A tennis ball of mass 0.060 kg is moving normally towards a racket with an initial velocity of